Unit-Cap Signed-Integer Selector Obstruction

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Abstract

RH-257 proves that integer exponents are necessary for a single-valued finite determinant quotient. We audit the first bounded signed lattice by assigning each complete RH-254 shell a weight in $\{-1, 0, 1\}$. Mixed-integer linear optimization gives zero anchor passes at all 32 endpoints. Optimal weighted distances range from 0.106074 to 0.353490, which is 7.98–93.48 times the local tolerance. The optimizations cover 39,417,456,084,975,216 lattice points in aggregate with zero reported MIP gap. This is a unit-cap obstruction only; larger caps and operator realization remain open.

1 The necessary integer lattice

For one endpoint let $v_j \in \mathbb{R}^{11}$ be the order-2–12 power vector of the jth conjugate-complete shell, and let d be the anchored target difference. RH-257 shows why a continuous signed weight is not enough: noninteger exponents create monodromy [3]. We therefore define the first integer relaxation

$$\mathcal{L}_1 = \left\{ \sum_{j=1}^{J} w_j v_j : w_j \in \{-1, 0, 1\} \right\}. \tag{1}$$

Every vector in $\mathcal{L}_1$ has a single-valued rational product at the level of formal shell factors. This remains only a necessary condition: negative or repeated shell multiplicities must still be realized by an actual quotient or superdeterminant.

Proposition 1.1 (MILP formulation)**.** *The distance*

$$\min_{w \in \{-1,0,1\}^J} \sum_{n=2}^{12} \frac{|d_n - (Vw)_n|}{n} \tag{2}$$

is an exact-integrality mixed-integer linear program with J integer variables and eleven nonnegative residual variables.

Proof. Introduce $u_n \geq 0$ and the two linear inequalities $d_n - (Vw)_n \leq u_n$ and $(Vw)_n - d_n \leq u_n$. Minimize $\sum_n u_n/n$ under integer bounds $-1 \leq w_j \leq 1$. □

2 Finite audit

The shell counts range from 20 to 34, so one endpoint contains 3^J lattice points. The optimizer searches this class implicitly rather than enumerating every vector. All 32 solves report zero MIP gap.

quantity	value
endpoints	32
aggregate lattice points	39,417,456,084,975,216
anchor passes	0
distance range	0.106074–0.353490
distance/tolerance range	7.98–93.48
minimum failure margin	0.103574
integrality-gap range	0.106074–0.353490
nonzero weights in an optimum	20–30
maximum branch-and-bound nodes	368
maximum reported MIP gap	0

Corollary 2.1 (Scoped unit-cap obstruction)**.** *No margin-32 complete-shell weight vector with entries in* $\{-1, 0, 1\}$ *reaches the deterministic anchor within the archived endpoint tolerance.*

Proof. At every endpoint the global MILP objective exceeds the local tolerance. The reported exact-integrality gap is zero. □

3 Boundary and route decision

The theorem does not exclude weights of magnitude two or larger. Such a fit would still not prove that the noisy operator contains the required signed copies. Rather than escalating an increasingly artificial multiplicity cap, the next paper returns to the exact orthogonal quotient and its block powers, where cancellation is operator-derived rather than fitted [1, 2].

Gates A–E remain false/open. No Hilbert–Polya operator, Riemann-zero identification, zeta-divisor equality, or RH implication is claimed.

References

[1] Bin Wang. Orthogonal quotient superloop compression. *Preprint*, 2026. RH-245 preprint.

[2] Bin Wang. Block-power quotient envelope criterion. *Preprint*, 2026. RH-246 preprint.

[3] Bin Wang. Monodromy integrality barrier for signed moment fits. *Preprint*, 2026. RH-257 preprint.